

## cwltool Conformance Tests Results

| Version    | Mode       | Passed | Failed | Skipped | Success (%) |
|------------|------------|--------|--------|---------|-------------|
| CWL v1.0.2 | login-node | 197    | 0      | 0       | 100.0       |
| CWL v1.0.2 | slurm-node | 197    | 0      | 0       | 100.0       |
| CWL v1.1.0 | login-node | 252    | 1      | 0       | 99.6        |
| CWL v1.1.0 | slurm-node | 252    | 1      | 0       | 99.6        |
| CWL v1.2.1 | login-node | 377    | 1      | 0       | 99.7        |
| CWL v1.2.1 | slurm-node | 377    | 1      | 0       | 99.7        |

Table 1: FinisTerae III conformance results using cwltool across CWL versions.

| Version    | Mode       | Passed | Failed | Skipped | Success (%) |
|------------|------------|--------|--------|---------|-------------|
| CWL v1.0.2 | login-node | 197    | 0      | 0       | 100.0       |
| CWL v1.0.2 | slurm-node | 197    | 0      | 0       | 100.0       |
| CWL v1.1.0 | login-node | 252    | 1      | 0       | 99.6        |
| CWL v1.1.0 | slurm-node | 252    | 1      | 0       | 99.6        |
| CWL v1.2.1 | login-node | 377    | 1      | 0       | 99.7        |
| CWL v1.2.1 | slurm-node | 377    | 1      | 0       | 99.7        |

Table 2: LUMI conformance results using cwltool across CWL versions.

| Version    | Mode       | Passed | Failed | Skipped | Success (%) |
|------------|------------|--------|--------|---------|-------------|
| CWL v1.0.2 | login-node | 197    | 0      | 0       | 100.0       |
| CWL v1.0.2 | slurm-node | 197    | 0      | 0       | 100.0       |
| CWL v1.1.0 | login-node | 252    | 1      | 0       | 99.6        |
| CWL v1.1.0 | slurm-node | 251    | 2      | 0       | 99.2        |
| CWL v1.2.1 | login-node | 377    | 1      | 0       | 99.7        |
| CWL v1.2.1 | slurm-node | 376    | 2      | 0       | 99.5        |

Table 3: MareNostrum 5 conformance results using cwltool across CWL versions.

## Toil Conformance Tests Results

| Version    | Mode        | Passed | Failed | Skipped | Success (%) |
|------------|-------------|--------|--------|---------|-------------|
| CWL v1.0.2 | login-node  | 197    | 0      | 0       | 100.0       |
| CWL v1.0.2 | slurm-node  | 197    | 0      | 0       | 100.0       |
| CWL v1.0.2 | slurm-batch | 191    | 6      | 0       | 97.0        |
| CWL v1.1.0 | login-node  | 250    | 1      | 2       | 98.8        |
| CWL v1.1.0 | slurm-node  | 250    | 1      | 2       | 98.8        |
| CWL v1.1.0 | slurm-batch | 246    | 7      | 0       | 97.2        |
| CWL v1.2.1 | login-node  | 375    | 1      | 2       | 99.2        |
| CWL v1.2.1 | slurm-node  | 375    | 1      | 2       | 99.2        |
| CWL v1.2.1 | slurm-batch | 371    | 7      | 0       | 98.1        |

Table 4: FinisTerae III conformance results using Toil across CWL versions.

| Version    | Mode        | Passed | Failed | Skipped | Success (%) |
|------------|-------------|--------|--------|---------|-------------|
| CWL v1.0.2 | login-node  | 195    | 2      | 0       | 99.0        |
| CWL v1.0.2 | slurm-node  | 195    | 2      | 0       | 99.0        |
| CWL v1.0.2 | slurm-batch | 48     | 149    | 0       | 24.4        |
| CWL v1.1.0 | login-node  | 247    | 4      | 2       | 97.6        |
| CWL v1.1.0 | slurm-node  | 249    | 2      | 2       | 98.4        |
| CWL v1.1.0 | slurm-batch | 90     | 163    | 0       | 35.6        |
| CWL v1.2.1 | login-node  | 374    | 2      | 2       | 98.9        |
| CWL v1.2.1 | slurm-node  | 373    | 3      | 2       | 98.7        |
| CWL v1.2.1 | slurm-batch | 176    | 202    | 0       | 46.6        |

Table 5: LUMI conformance results using Toil across CWL versions.

| Version    | Mode        | Passed | Failed | Skipped | Success (%) |
|------------|-------------|--------|--------|---------|-------------|
| CWL v1.0.2 | login-node  | 197    | 0      | 0       | 100.0       |
| CWL v1.0.2 | slurm-node  | 157    | 33     | 7       | 79.7        |
| CWL v1.0.2 | slurm-batch | 70     | 124    | 3       | 35.5        |
| CWL v1.1.0 | login-node  | 250    | 1      | 2       | 98.8        |
| CWL v1.1.0 | slurm-node  | 206    | 38     | 9       | 81.4        |
| CWL v1.1.0 | slurm-batch | 124    | 124    | 5       | 49.0        |
| CWL v1.2.1 | login-node  | 373    | 3      | 2       | 98.7        |
| CWL v1.2.1 | slurm-node  | 320    | 43     | 15      | 84.7        |
| CWL v1.2.1 | slurm-batch | 171    | 203    | 4       | 45.2        |

Table 6: MareNostrum 5 conformance results using Toil across CWL versions.

## StreamFlow Conformance Tests Results

| Version                                                             | Mode       | Passed                                                 | Failed | Skipped | Success (%) |
|---------------------------------------------------------------------|------------|--------------------------------------------------------|--------|---------|-------------|
| CWL v1.0.2                                                          | login-node | 162                                                    | 18     | 1       | 89.5        |
| CWL v1.0.2                                                          | slurm-node | 170                                                    | 26     | 1       | 86.3        |
| <i>CWL v1.1.0: conformance suite not executed (launch failure).</i> |            |                                                        |        |         |             |
| CWL v1.2.1                                                          | login-node | 314                                                    | 31     | 2       | 90.5        |
| CWL v1.2.1                                                          | slurm-node | <i>conformance suite not executed (launch failure)</i> |        |         |             |

Table 7: FinisTerae III conformance results using StreamFlow across CWL versions.

| Version                                                             | Mode       | Passed                                                 | Failed | Skipped | Success (%) |
|---------------------------------------------------------------------|------------|--------------------------------------------------------|--------|---------|-------------|
| CWL v1.0.2                                                          | login-node | 196                                                    | 0      | 1       | 99.5        |
| CWL v1.0.2                                                          | slurm-node | 196                                                    | 0      | 1       | 99.5        |
| <i>CWL v1.1.0: conformance suite not executed (launch failure).</i> |            |                                                        |        |         |             |
| CWL v1.2.1                                                          | login-node | 375                                                    | 1      | 2       | 99.2        |
| CWL v1.2.1                                                          | slurm-node | <i>conformance suite not executed (launch failure)</i> |        |         |             |

Table 8: LUMI conformance results using StreamFlow across CWL versions.

| Version                                                             | Mode       | Passed | Failed | Skipped | Success (%) |
|---------------------------------------------------------------------|------------|--------|--------|---------|-------------|
| CWL v1.0.2                                                          | login-node | 157    | 39     | 1       | 79.7        |
| CWL v1.0.2                                                          | slurm-node | 157    | 39     | 1       | 79.7        |
| <i>CWL v1.1.0: conformance suite not executed (launch failure).</i> |            |        |        |         |             |
| CWL v1.2.1                                                          | login-node | 325    | 51     | 2       | 86.0        |
| CWL v1.2.1                                                          | slurm-node | 325    | 51     | 2       | 86.0        |

Table 9: MareNostrum 5 conformance results using StreamFlow across CWL versions.